Oxford College of Emory University

Chapter 14, Part 1

Money, Liquidity, and Banking

Guilherme Paiva Pinto

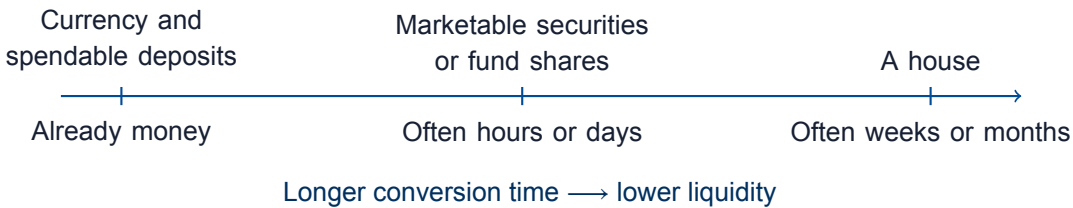
October 27, 2026

Roadmap

1. Explain money's functions, liquidity, and the measures M1 and M2.
2. Describe what banks do and read a simple T-account.
3. Explain bank runs using Silicon Valley Bank as an example.
4. Explain bank safeguards, moral hazard, and shadow banking.

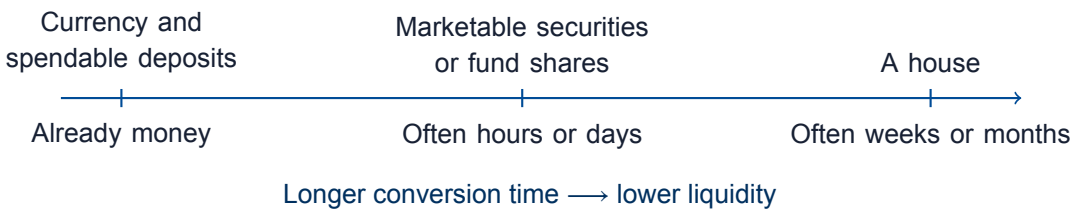
Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.



Money and Liquidity

- **Money**: what people readily accept as payment for goods and services.
- **Liquidity**: how quickly something can become spendable money, with little cost or loss of value.



The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.

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The Money Supply: M1 and M2

- **Money supply**: the total amount of currency and bank balances counted as money.
- **M1, the narrower measure**: public currency, demand deposits, and other liquid deposits, including savings deposits.
- **M2, the broader measure**: M1 plus small time deposits and retail money market mutual fund balances.

M2 = M1 + additional near-money. Near-money is readily convertible into spendable balances.

Source: Federal Reserve Board, H.6 definitions.

The Functions of Money

- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.

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- **Medium of exchange**: an accepted way to pay for goods and services.
  - Pay for groceries without a double coincidence of wants.
- **Unit of account**: a common unit for stating and comparing values.
  - Compare a $5 smoothie with a $10 sandwich.
- **Store of value**: a way to carry purchasing power into the future.
  - Save today's earnings for next month's groceries. Inflation can erode their value.

Types of Money

A farmer brings produce to market. Compare three ways to pay.



Producing crops: Bruegel, *The Harvesters*, 1565



Trading: Van Ruysdael, *Market by the Seashore*, 1637

Commodity money

Pay with a portion of the crop accepted as money. It also has value as food.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

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Pay with a warehouse receipt redeemable for a specified quantity of grain.

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Fiat money
Pay with officially accepted money that promises no commodity redemption.

Images: The Met, 19.164 and The Met, 60.55.4; both CC0.

What Banks Do

- Banks **accept deposits** and provide payment services.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.

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What Banks Do

- Banks **accept deposits** and provide payment services.
- They keep part of their funding in **reserves** and lend a large portion or buy investments such as bonds.
- Deposits can be withdrawn quickly; many loans and investments repay over years.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Bank Reserves

- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.

Source: Federal Reserve Board, The Fed Explained.
Source: Federal Reserve Board, Reserve Requirements.

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- Reserves let a bank meet withdrawals and settle payments to other banks.

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- **Bank reserves**: cash in bank vaults plus the bank's deposits at the Federal Reserve.
- Reserves let a bank meet withdrawals and settle payments to other banks.
- **Reserve ratio**: bank reserves divided by the bank's deposits.

Source: Federal Reserve Board, The Fed Explained.
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T-Accounts

A **T-account** shows **what a bank has on the left** and **what it owes on the right**.

What the bank has		What the bank owes	
Loans	$1,200,000	Deposits	$1,000,000
Reserves	$100,000		

Total on the left: $1,300,000. Deposits: $1,000,000. The $300,000 difference is owners' money, which absorbs losses first.

Numerical example: Krugman and Wells, Macroeconomics, 7th ed., Chapter 14.

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Reserve ratio $= \frac{100{,}000}{1{,}000{,}000} = 10\%.$

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Money and Banking Checkpoint

A customer deposits $100 of currency into a checking account.

What changes for the bank and for public currency + deposits?

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What changes for the bank and for public currency + deposits?

Reserves +$100; deposits +$100.

Public currency −$100, so currency + deposits is unchanged.

Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.

Source: Federal Reserve Board, SVB review, April 2023.

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- The bank has lent or invested much of that money. It cannot get it all back immediately.

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Bank Runs

- A **bank run** happens when many customers rush to take out their money because they fear the bank may fail.
- The bank has lent or invested much of that money. It cannot get it all back immediately.
- Selling investments quickly can mean losses. This can make more customers want to leave.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: March 2023

- **2022**: higher interest rates made SVB's older, long-term bonds worth less.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

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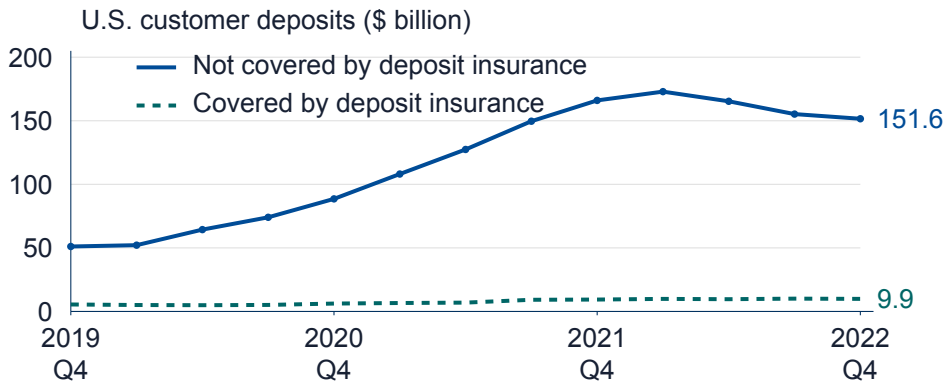
Silicon Valley Bank: March 2023

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- **March 8–9**: SVB announced losses. Worried customers withdrew more than $40 billion the next day.
- **March 10**: the bank could not meet the rush of withdrawals. Regulators closed it.

"Silicon Valley Bank shut by California regulator"
Reuters, March 10, 2023.

Source: Federal Reserve Board, SVB review, April 2023.

Silicon Valley Bank: Customer Deposits



Original graph. Data: Federal Reserve Board, April 2023 SVB review, Figure 7 data; Call Report.

Bank Regulation

- **Deposit insurance**: the FDIC, created in 1933, protects eligible deposits up to $250,000 per depositor, insured bank, and ownership category.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

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- **Interest on reserve balances (IORB)**: the Fed pays interest on money banks keep in their accounts at the Fed.
- **Capital requirements**: owners must put in their own money to absorb losses first.

Sources: FDIC history; deposit coverage; Federal Reserve: discount window, IORB, capital.

Moral Hazard

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- Customers may watch risks less closely if their deposits are insured. Owners may gain from risky bets while others bear some losses.
- **Capital requirements** make owners risk more of their own money; bank oversight helps enforce the rules.

Source: Federal Reserve Board, regulatory capital.

Shadow Banking

- **Shadow banking**: lending and borrowing through firms that are not ordinary deposit banks.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

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- Examples: **investment banks** help companies raise money; **money market funds** pool savings for short-term lending.

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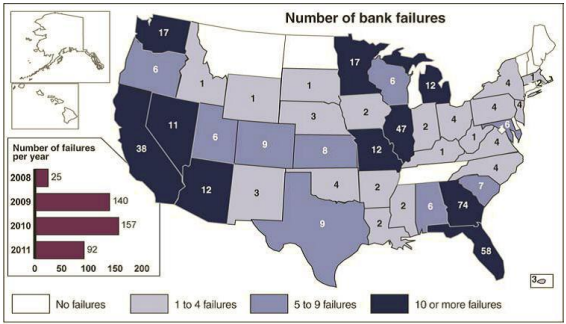
The 2008 crisis

- September 15: **Lehman Brothers** went bankrupt.
- September 16: losses at **Reserve Primary Fund** led investors to demand their money back.

Source: Financial Stability Board, Non-Bank Financial Intermediation.

Source: Federal Reserve Board, crisis institutions and funding runs (2009).

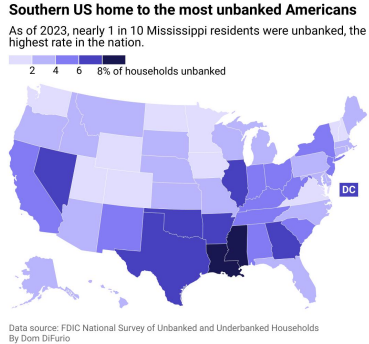
Bank Failures and Unbanked Households



Source: GAO analysis of FDIC data: Map Resources (map).

Map: GAO analysis of FDIC data, 2008–2011.

Reference: FDIC (2020), Staff Studies 2020–05, Table 1.



FDIC (2024), 2023 Household Survey, state estimates.

Visual: Dom DiFurio. Measure: share of households.

Oxford College of Emory University

Money Creation and Central Banks

Guilherme Paiva Pinto

October 29, 2026

Roadmap

1. Explain how banks create money.
2. Calculate and interpret the simple money multiplier.
3. Understand how central banks work and identify their policy tools.
4. Understand how lessons from financial crises shaped monetary policymaking.

Chapter 14, Part 2 · Section 1

Explain how banks create money.

Trace deposits, loans and payments, distinguishing changes in the money supply from changes in the monetary base.

Determining the Money Supply

Money supply = public currency + checkable deposits

Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing			

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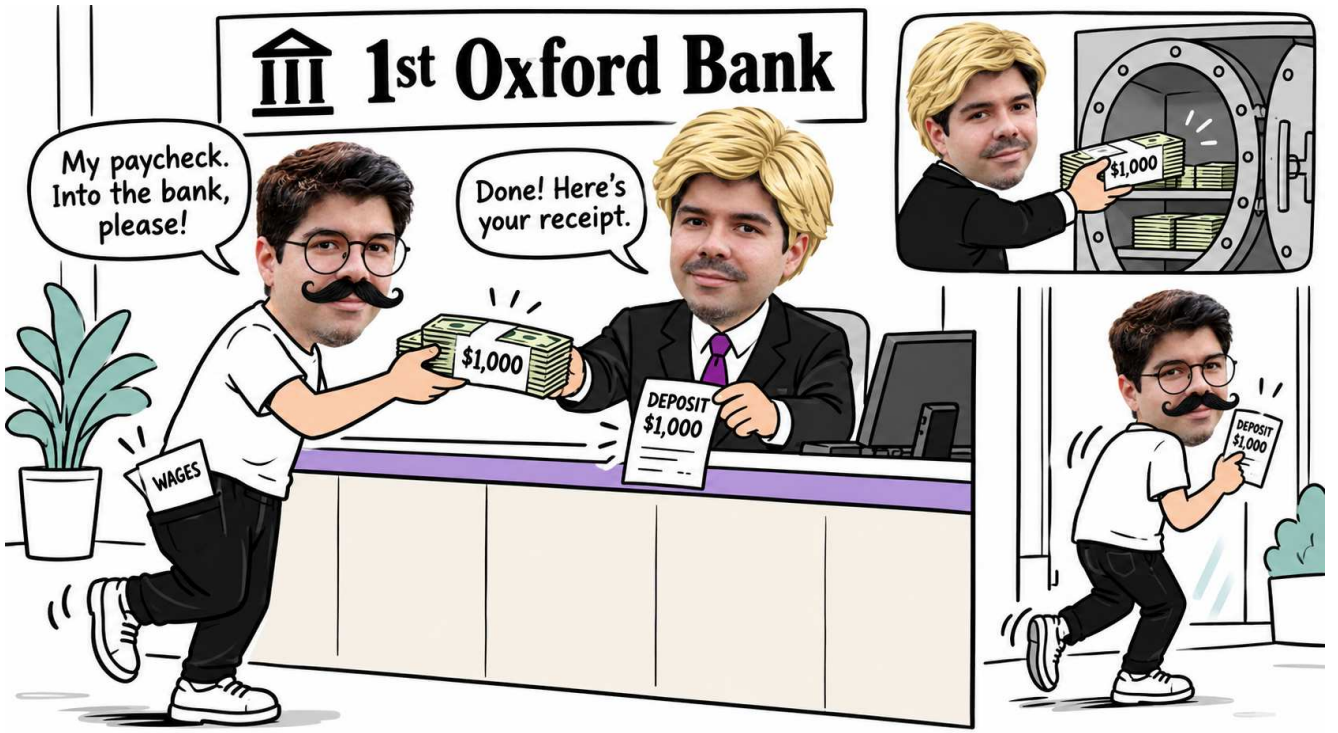
Simplified measure: M = C + D

The same $1,000	Currency (C)	Deposits (D)	Money (M)
Before depositing	$1,000	$0	$1,000
After depositing	$0	$1,000	$1,000

The deposit changes the form of money, not its total.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



The cash deposit changes the form of $1,000, not the money total.

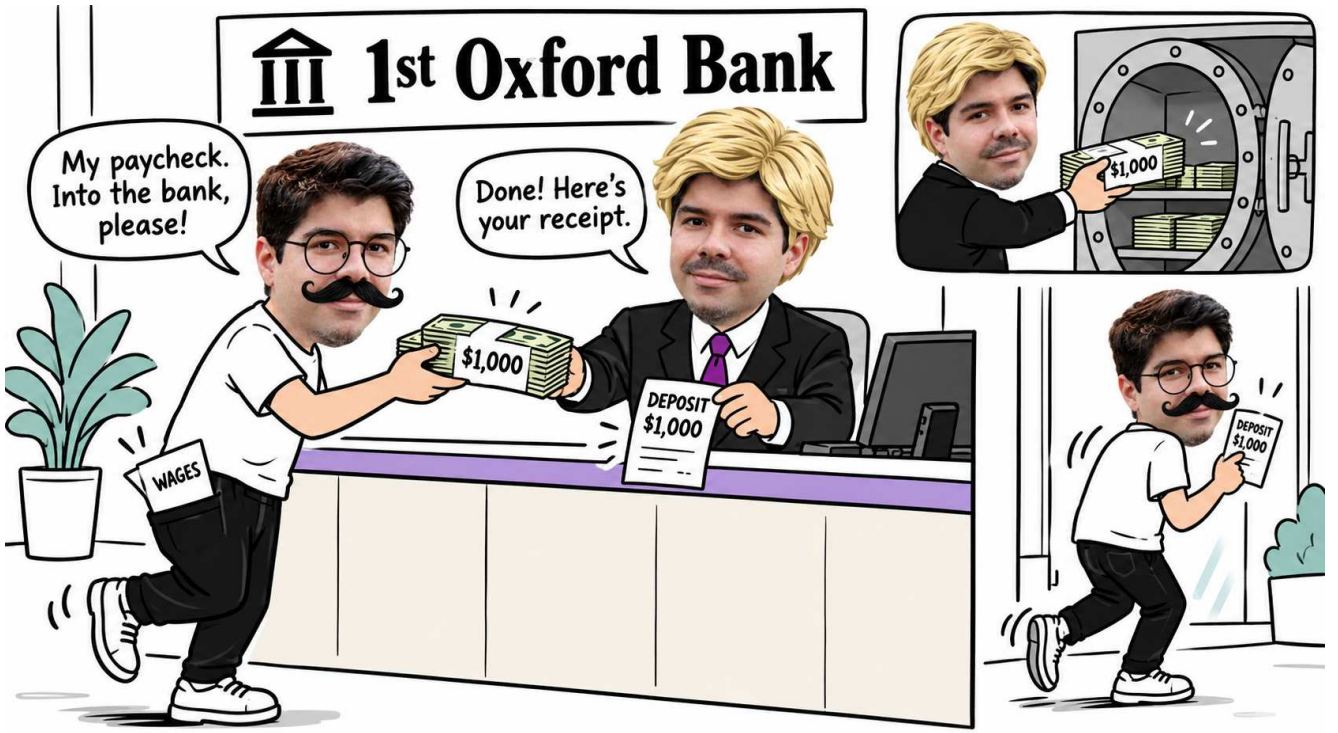
How banks create money

Stage	Currency in circulation	Checkable bank deposits	Money supply
First stage Cash held by the public	$1,000	$0	$1,000
Second stage First deposit → $900 loan and payment			
Third stage $900 redeposit → $810 new loan			

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

How Banks Create Money

Initial cash: $1,000 · Reserve ratio: rr = 10%



First bank: $100 required reserves; $900 available to lend.

How banks create money

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How Banks Create Money

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The first bank lends $900 in cash.

How banks create money

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The borrower pays the same $900 to the supplier.

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The next bank receives $900, keeps $90 and lends $810.

The third stage includes the **next $810 loan**.

Numerical stages: Krugman & Wells, 7th ed., Chapter 14, Table 1. Each row includes the completed loan stage.

Chapter 14, Part 2 · Section 2

Calculate and interpret the simple money multiplier.

Determine total deposit expansion and explain the assumptions that limit the model’s application to actual banking.

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

rr = 10% · Each row follows lending and payment to the next bank.

Bank	Deposits	Reserves kept	New loan
A	$1,000		
B			
C			
…	…	…	…

Reserves and Repeated Deposit Creation

Excess reserves = reserves − required reserves.

At Bank A: $1,000 − (0.10 × $1,000) = **$900**.

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B	$900	$90	$810
C	$810	$81	$729
…	…	…	…

The Simple Money Multiplier

Initial cash deposit: \$1,000 · rr = 0.10

$$D = 1{,}000 + 1{,}000(1 - rr) + 1{,}000(1 - rr)^2 + \ldots$$

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$10,000 total deposits includes the original $1,000.

$9,000 additional deposits are created through lending.

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\$9,000 additional deposits are created through lending.

Under these assumptions, \$1 of reserves supports \$10 of deposits.

Why Reality Is More Complicated

Full redeposit. People deposit the payments they receive.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Why Reality Is More Complicated

Full redeposit. People deposit the payments they receive.

Only required reserves are retained. Banks keep rr × deposits and lend the remaining amount.

At rr = 10%: a $900 deposit → $90 reserves + $810 loan.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Why Reality Is More Complicated

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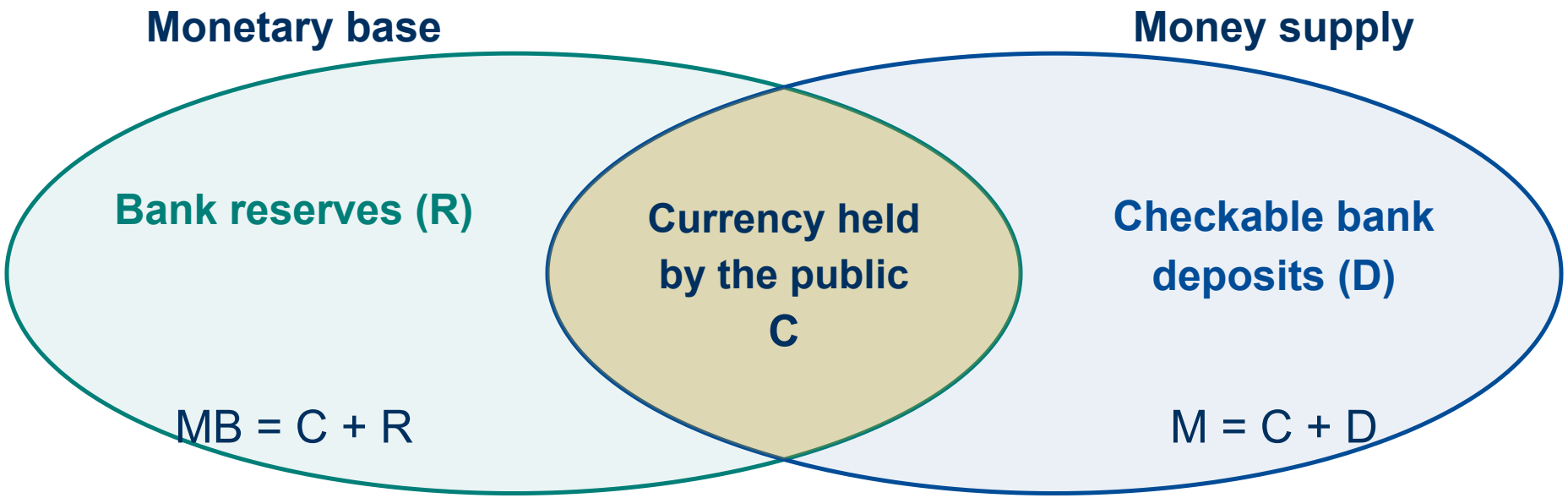
More cash held by the public or more excess reserves
means less deposit expansion.

1/rr is the model's simple deposit multiplier.

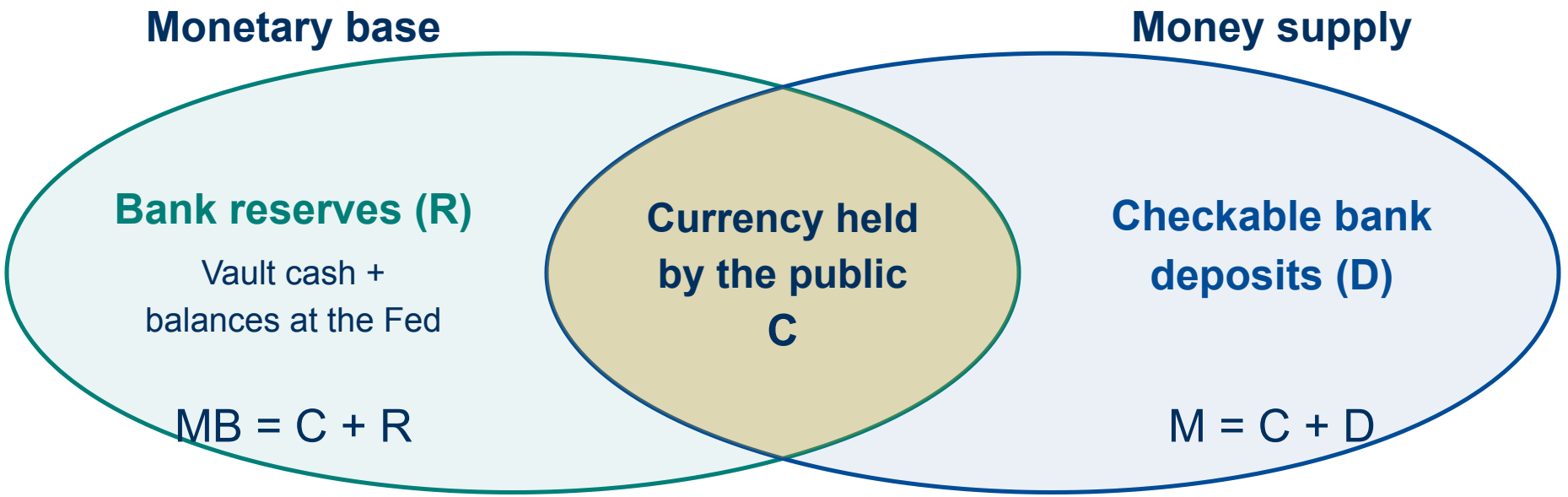
rr = 10% is our example; the U.S. statutory requirement is 0%.

Bank of England, Money Creation (2014) · Federal Reserve, Reserve Requirements

Monetary Base and Money Supply

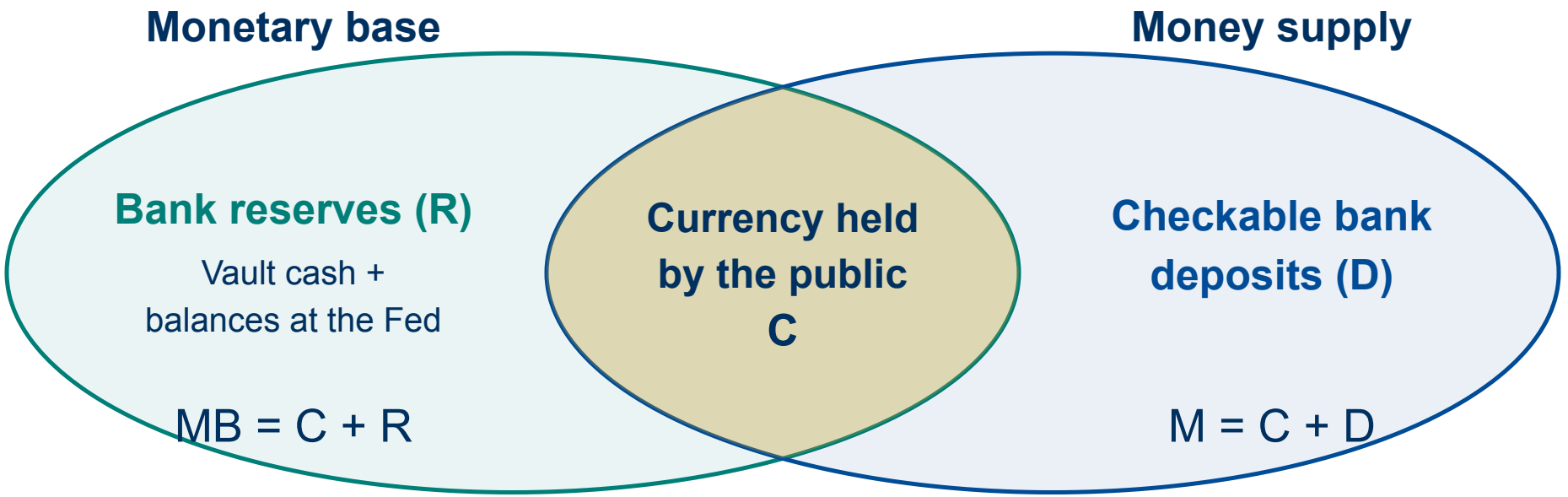


Monetary Base and Money Supply



Public currency is the overlap. Bank reserves and public deposits are different components.

Monetary Base and Money Supply



Public currency is the overlap. Bank reserves and public deposits are different components.

$$\text{Money multiplier} = \frac{M}{MB}$$

Next: how Fed operations change reserves and the monetary base.

Money Creation Checkpoint

$1,000 of existing public cash · rr = 10% · Full lending and redeposit

1. Cash becomes a deposit.

Does the money supply rise immediately?

2. Lending repeats across banks.

What are total deposits and the net increase in money?

3. Banks make loans.

Does lending itself create monetary base?

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Does lending itself create monetary base?

No. It creates deposits; it does not create base money.

Chapter 14, Part 2 · Section 3

Understand how central banks work and identify their policy tools.

Explain the roles of the discount rate, interest on reserve balances and open-market operations.

The Federal Reserve

The Federal Reserve (the Fed) is the U.S. central bank.

Its monetary-policy goals: maximum employment and stable prices.

Federal Reserve, System Structure

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Monetary policy: influence borrowing and spending to help reach those goals.

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Its monetary-policy goals: maximum employment and stable prices.

Monetary policy: influence borrowing and spending to help reach those goals.

The tools we will study:

- Loans from the Fed to banks
- Interest paid on banks' balances at the Fed
- Purchases and sales of bonds

Federal Reserve, System Structure

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

12 Reserve Banks. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 governors · Washington, D.C.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

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10. Kansas City
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12. San Francisco

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Board of Governors

7 governors · Washington, D.C.

Each governor:
The U.S. President nominates;
the U.S. Senate confirms.

Full term: 14 years.
One term ends every two years.

Chair: chosen from the governors by the same process, for four years.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

The Federal Reserve System



1. Boston
2. New York
3. Philadelphia
4. Cleveland
5. Richmond
6. Atlanta
7. Chicago
8. St. Louis
9. Minneapolis
10. Kansas City
11. Dallas
12. San Francisco

12 Reserve Banks. San Francisco also serves Alaska and Hawaii.

Board of Governors

7 governors · Washington, D.C.

Each governor:
The U.S. President nominates;
the U.S. Senate confirms.

Full term: 14 years.
One term ends every two years.

Chair: chosen from the governors by the same process, for four years.

Why regional banks? Share responsibility and bring local economic conditions into national decisions.

Federal Reserve, System Structure · Federal Reserve, Governor Selection · Map: Natural Earth

The Federal Open Market Committee

FOMC = the Fed’s monetary-policy committee.

7	+	1	+	4
Board governor seats		New York Fed president		Other regional presidents
Permanent votes		Permanent vote		One-year voting turns

Federal Reserve, FOMC

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Directs bond purchases and changes in the Fed’s bond holdings.

For our activity: the statement explains the decision and goals; the implementation note gives the operating steps.

The Board of Governors sets reserve interest and approves discount rates.

Federal Reserve, FOMC

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

New York Fed, Monetary Policy Implementation

What the Fed Does

Monetary policy · financial stability and supervision · payments

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FFR: the federal funds rate—the interest rate in this market.

New York Fed, Monetary Policy Implementation

What the Fed Does

Monetary policy · financial stability and supervision · payments

The federal funds market

Overnight, unsecured lending of balances at the Fed between banks and other eligible institutions.

FFR: the federal funds rate—the interest rate in this market.

Policy tool	What changes
Discount rate	Cost of borrowing from the Fed
Interest on reserve balances	Return on eligible balances at the Fed
Open-market operations	Securities and bank reserve balances

Next week: interest rates, spending, and aggregate demand.

New York Fed, Monetary Policy Implementation

The Discount Window and Reserve Interest

Discount window

The Fed lends to eligible banks against collateral.

Interest on reserve balances

IORB pays banks on eligible balances held at the Fed.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

The Discount Window and Reserve Interest

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Lower discount rate:
Cheaper Fed funding; borrowing becomes more attractive.

Higher discount rate:
Costlier Fed funding; borrowing becomes less attractive.

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A reservation rate:
a safe alternative return to lending.

Higher IORB raises that benchmark; lower IORB reduces it.

IORB influences lending rates; it is not a fixed minimum rate on every loan.

Federal Reserve, Discount Window · Federal Reserve, IORB FAQ

Open-Market Purchases

The Fed buys $1,000 of government bonds from a commercial bank.



Purchase: reserves enter the bank.

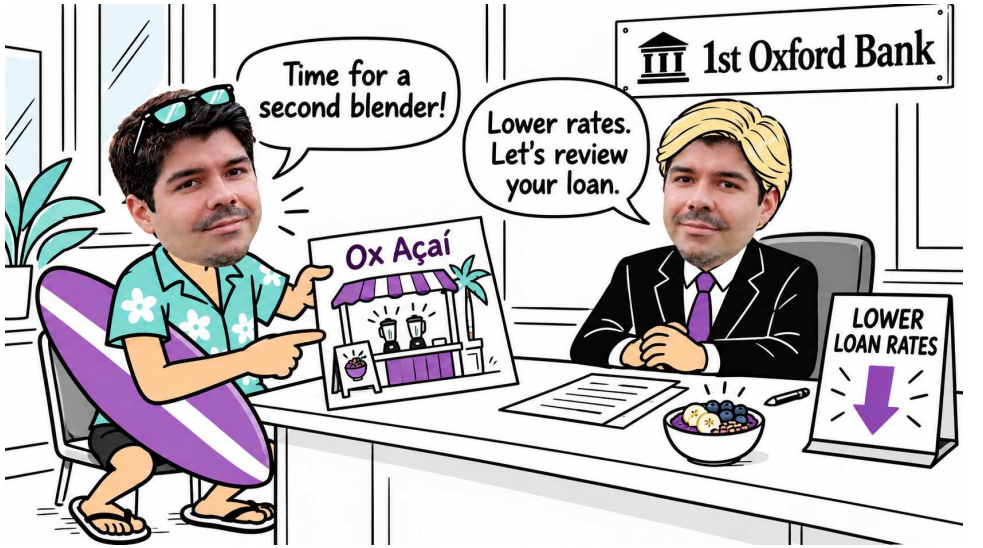
New York Fed, Monetary Policy Implementation

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Easier financing can support borrowing.

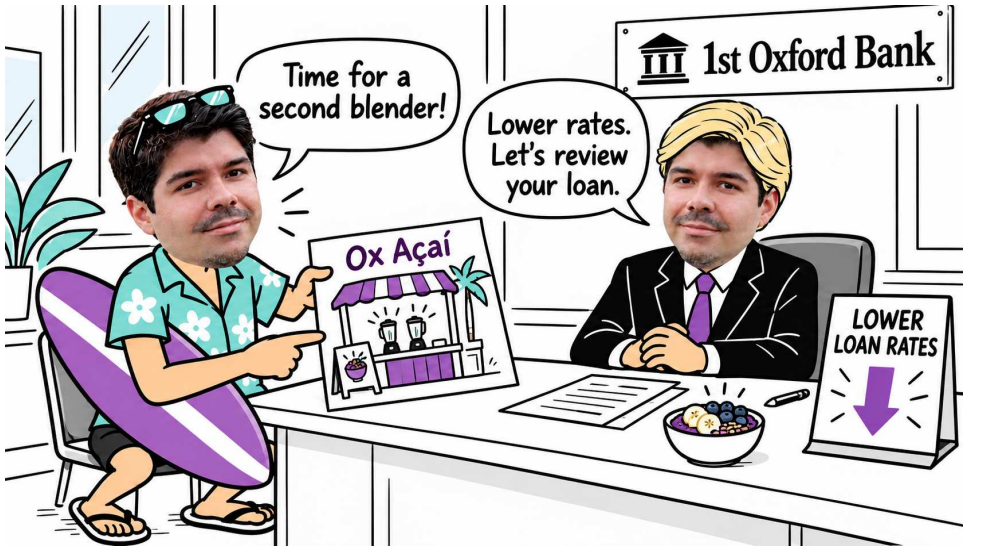
New York Fed, Monetary Policy Implementation

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Bank reserves +$1,000 · Monetary base +$1,000

New York Fed, Monetary Policy Implementation

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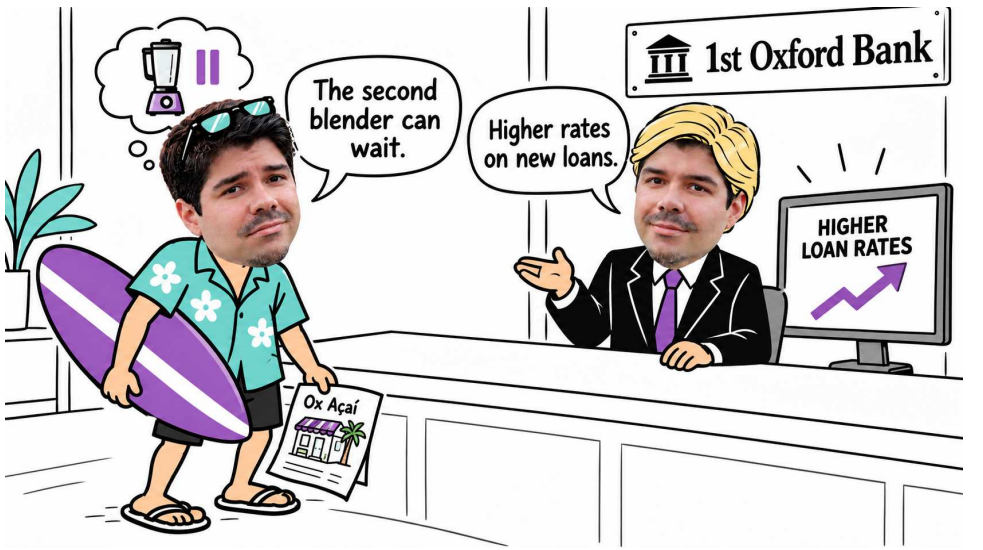
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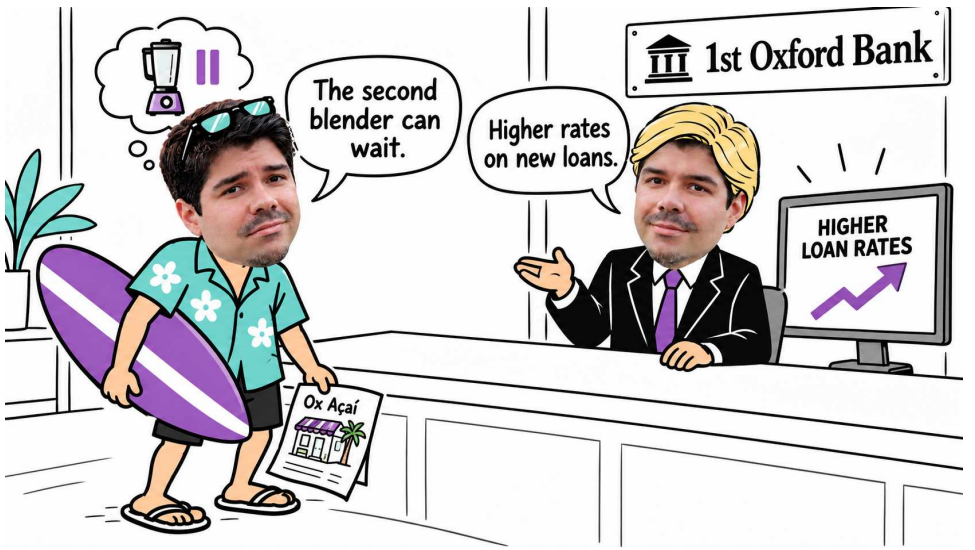
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New York Fed, Monetary Policy Implementation

The ECB and the Federal Reserve

ECB = European Central Bank · Eurosystem = ECB + euro-area national central banks.

	Federal Reserve	ECB / Eurosystem
Central board	7 Board governor seats	6 Executive Board members
Policy committee		
Operating network		
Main policy goals		

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

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Main policy goals	Maximum employment; stable prices	Price stability is primary

Both use policy rates, lending facilities, and asset operations.

The ECB’s deposit facility offers a comparable remuneration benchmark.

ECB, Governing Council · ECB, Eurosystem · ECB, Monetary Policy · ECB, Standing Facilities · Federal Reserve, System Structure

Chapter 14, Part 2 · Section 4

Understand how lessons from financial crises shaped monetary policymaking.

Connect historical crises to changes in central banks’ responsibilities and policy approaches.

Bank Runs and the Creation of the Fed

1907–1913: panic and the Fed's creation

A failed bet on copper-company shares helped trigger panic in 1907. People rushed to withdraw deposits, and banks struggled to find cash. The Fed was created on **December 23, 1913** to help meet these cash needs.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

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1930–1934: Great Depression bank runs

New runs helped force thousands of banks to close. Lending collapsed, and the Fed failed to stop the crisis. Federal deposit insurance was created in **1933** and began in **1934**.

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Policy lesson: The Fed provides emergency bank loans; the FDIC protects eligible deposits.

Fed History: Panic of 1907 · Fed History: Federal Reserve Act · Fed History: Great Depression

Stagflation and High Interest Rates

1970s: stagflation

High inflation came with weak growth and high unemployment. Oil-price shocks made both problems worse.

Fed History: Great Inflation · Fed History: Recession of 1981–82 · BLS: Annual CPI Inflation

Stagflation and High Interest Rates

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1979–1981: the Fed raises rates

Under Paul Volcker, short-term interest rates rose to nearly **20%**. Borrowing became much more expensive, slowing spending.

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1981–1983: costs and results

A deep recession pushed unemployment to nearly **11%** in 1982. Annual inflation fell from **13.5% in 1980** to **3.2% in 1983**.

Policy lesson: Lower inflation came with weaker spending and serious job losses.

Fed History: Great Inflation · Fed History: Recession of 1981–82 · BLS: Annual CPI Inflation

The 2008 Crisis and the Fed’s Response

The problem: Investment firms and funds relied on short-term borrowing. Home-loan losses made their lenders pull their money out—a run outside ordinary banks.

2007–2008: discount window and emergency loans

Cheaper, longer loans for banks. Separate emergency programs also lent to some investment firms.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008

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2008–2009: larger open-market purchases

Bought large amounts of longer-term bonds, including bonds backed by home loans, to lower borrowing costs and support spending.

Policy lesson: The crisis brought broader Fed lending, larger bond purchases and a new tool for guiding rates.

Federal Reserve: Emergency Lending, March 2008 · Fed History: Interest on Reserves · Federal Reserve: Bond Purchases, November 2008

Reference Definitions and Examples

Money, income, wealth, and bank accounting.

Money, Income, and Net Wealth

Concept	Meaning	Example
Money	Readily spendable assets	$400 in currency and deposits
Income	Receipts over a period	$900 earned this month
Net wealth	Assets minus liabilities	$5,000 − $2,000 = $3,000

A Smoothie Company Balance Sheet

Assets	Liabilities and equity
Cash and inventory	Loans and other obligations
Equipment; a building if owned	Equity: the owners' residual claim

Assets = liabilities + equity.

Liquidity and Solvency

Issue	Question	Possible problem
Liquidity	Can the bank pay now?	Assets cannot all be sold quickly at little cost.
Solvency	Do assets cover liabilities?	Losses can exhaust equity.

Forced sales can turn liquidity pressure into losses.

A Cash Deposit

Assets ($)	Liabilities and equity ($)
Reserves: 200	Deposits: 1,000
Loans: 900	Equity: 100
Total: 1,100	Total: 1,100

A customer deposits $200 of currency held outside banks.

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Reserves and deposits each rise by $200.

Public currency falls by $200; currency + deposits is unchanged.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.

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A New Bank Loan

- The bank approves a $120 loan and credits the borrower's deposit.
- Loans and deposits each rise by $120 at origination.
- The borrower gains a deposit and owes a debt; net wealth does not rise by $120.

Source: McLeay, Radia and Thomas, Bank of England (2014), conceptual reference.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.

Source: Federal Reserve Board, H.6 definitions.

Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.

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Reserves and the Monetary Base

- **Public currency**: notes and coins outside banks.
- **Bank reserves**: vault cash and balances at the Fed.
- **Monetary base**: public currency plus bank reserves.

Source: Federal Reserve Board, H.6 definitions.